

Why Does So Many People Die of Heart Attack

For many decades, cardiovascular disease has remained the leading cause of death in the United States and elsewhere, in spite of amazing medical advances that have helped our ability to treat cardiovascular disease with life-saving devices, such as stents, implantable cardio converter defibrillators ,advances in cardiac surgery and advancement in medications.

We also understand now the importance of healthy diet, physical activity, good quality sleep and smoking cessation for cardiovascular health. In spite of that almost 900,000 people died in the United States from cardiovascular disease in 2024, which is more than the number of people killed by cancer and accidents combined.

At present, we have clear understanding of the risk factors for heart disease. Study published in 2025 showed that more than 99% of the people who had a heart attack, stroke, or heart failure had at least one cardiovascular risk factor. It is clear that cardiovascular disease is preventable.

Unfortunately, we failed to recognize and treat these risk factors earlier, such as hypertension, hyperlipidemia, diabetes, and kidney disease. We also know very well that most of the cardiovascular risk factors do not develop overnight and it takes often decades. However, when someone gets a heart attack or stroke, we often viewed that as a sudden or unexpected.

In fact, at present, we have tools to predict the likelihood of a cardiovascular event way before this happens. In 2023, American College of cardiology and American Heart Association came up with the PREVENT EQUATION (Predicting Risk of Cardiovascular Disease Events).

This calculator can predict risk of heart attack, stroke and heart failure over the next 10 or 30 years and this guidelines recommend physicians to use the PREVENT calculator for cardiovascular risk assessment for patient between the ages of 30 and 79 years.

Many of these risk factors are silent. Many people may not even know that their blood pressure is high, blood sugar levels are abnormal or cholesterol levels are elevated. If untreated, the underlying cardiovascular disease progresses relentlessly without your knowledge until a catastrophic event occurs .

Just like you start planning for your retirement decades in advance, your health requires the same early and proactive investment.

BENEFITS OF PREVENT CALCULATOR Risk estimation using the new calculator should begin earlier, at age 30 instead of 40. And for people in their 30s through 50s, it should include estimation of 30 year risk which this Calculator can do. For patients whose young age or female sex keep their 10 year risk low, knowing that the long-term risk is elevated can help to benefit lifestyle modifications.

PREVENT RISK CATEGORY RANGES Using the prevent calculator, the 10 year risk categories have been recalibrated - Low risk: less than 3%(previously less than 5%) - Borderline risk: 3% to less than 5% (previously 5% to less than 7.5%) - Intermediate risk: 5% less than 10% (previously 7.5% to less than 20%) - High risk: 10% or higher (previously 20% or higher) For primary prevention, the guidelines say lipid lowering therapy CAN BE considered for patients with borderline risk and SHOULD BE considered for those with intermediate risk

LDL CHOLESTEROL GOAL - Less than 100 mg/dL for patients without clinical cardiovascular disease who have low borderline, or intermediate risk based on PREVENT EQUATION - Less than 70 mg/dL for patients without clinical cardiovascular disease who have high risk based on PREVENT EQUATION and a small number of patients with clinical cardiovascular disease who do not have high risk - Less than 55 mg/dL for most patients with clinical cardiovascular disease.

It is also true that sometimes, risk of developing cardiovascular disease is not entirely within a person's control. Take for example, family history, ethnicity, social, economical, and environmental factors, access to healthcare.-these are all coming to play.

Testing Lp(a) once for everyone Elevated Lp(a) level is associated with increased cardiovascular risk. Patient with levels above 50 mg/dL have 1.4 - fold increased risk of cardiovascular disease and the risk doubles at values at or above 100 mg/dL and quadruplets at or values at or above 180 mg/dL. Therefore, guidelines now recommends, Lp(a) testing for all patients.

Repeating the test generally is not necessary because Lp(a) genetically determined and tend to remain stable overtime. Although Lp(a) lowering oral medications are not yet available, patient with elevated levels should have more intense LDL cholesterol lowering treatments and tighter control of additional risk factors.

Trials testing the effects of lowering Lp(a) levels on the heart attacks and strokes are expected to be reported later this year. This result will surely bring another update later on this year or next year.

Apo B TESTING ApoB may help to predict future cardiovascular events and whether or not there is a potential for genetic abnormality. A patient's ApoB level represents their total number atherogenic lipoprotein particles, which include LDL, very low density, lipoprotein, and Lp(a) -and it is not affected by fasting.

The guidelines suggests that apoB testing can help to identify residual Lipo protein-related risk in patient who have met their LDL cholesterol and non-HDL cholesterol goals. It can be particularly useful in those with cardiovascular disease, elevated triglycerides, type 2 diabetes or cardiovascular-kidney-metabolic syndrome

CORONARY ARTERY CALCIUM SCORING Coronary artery calcium typically emerges after age 40 in men, and after age 45 in women. So guideline suggests coronary artery calcium scoring -CAC scoring for intermediate risk or borderline risk patients with no prior cardiovascular disease.

This test can help in decision about initiating lipid lowering medications or whether someone needs lipid lowering therapy or not. Even incidental findings of coronary calcium on a general chest CT is also a useful information.

RISK ENHANCING FACTORS This will add to the general risk factors.

These risk enhancers include; - Female specific risks, including early menopause before the age of 45, polycystic ovary syndrome, adverse pregnancy outcomes such as hypertension during pregnancy, gestational diabetes, and preterm delivery - South Asian ethnicity like people from Indian subcontinent and - also Filipinos - Other factors include family, history of early coronary artery disease, kidney disease , metabolic syndrome, elevated Lp(a) level, persistently, elevated LDL cholesterol or triglyceride, elevated high sensitivity, CRP - History of inflammatory disease like rheumatoid arthritis, lupus, and HIV

HIGH TRIGLYCERIDES For patient with high triglycerides, consulting a dietitian is crucial. Although fibrates like fenofibrate or Icosapent ethyl (Vascepa) can be added, statins are still the mainstay of drug therapy. Non-pharmacological therapy, including weight reduction, diet, and regular exercise are all helpful.

INFORMATION ON DIETARY SUPPLEMENTS For the first time, the guidelines do not recommend the use of dietary supplements, such as non-prescription fish oil, to lower LDL cholesterol or triglyceride levels in people with atherosclerotic cardiovascular disease. According to the guidelines, there is evidence of absence of benefits with these supplements.

As usual, the above is for information purpose only and individuals should consult their own physician about their own situation.

Keshava Aithal

Dr Keshava Aithal

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